The formal complexity of Nanti stress

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Introduction

- Stress patterns across languages have been widely investigated in the subregular domain (Heinz, 2009; Heinz et al., 2011; Lambert, 2026)
  - Strictly Local (Heinz 2010), Strictly Piecewise (Heinz 2010), or their tier-based extensions (Heinz et al., 2011; Jardine & Heinz 2015)
- Attested phonotactic patterns can be described with these types of factors (subsequences, substrings) in **a propositional logic** (Lambert, 2026)
- Complicated stress patterns with **interacting multiple phonological factors** have not yet been fully investigated.

Nanti

- Nanti (a Kampa language in Peru) is a **complicated stress system** with multiple interacting phonological and rhythmic factors (Crowhurst & Michael 2005):
  - rhythm (iambic or trochaic)
  - strength $\in$ {syllable weight, vowel quality}
  - stress clash avoidance
  - non-finality
  - foot binarity
  - penultimate-sensitivity to primary stress

Overview

- I present a computational analysis of the Nanti stress system, using *the Language Toolkit* (Lambert 2024).
- The Nanti stress system is **complicated**: rough estimate of 2,000 states for the full 12-level machine.
- By the complexity metrics (state size, alphabet size, logical expressivity), it is **more complex** than other stress patterns.
- It is defined as **First-Order with precedence** but neither FO2($<$) nor FO($\triangleleft$), placing it above the propositional classes.
- This result has implications for both typology and learnability.

Roadmap

1. Stress pattern of Nanti
2. The construction of the finite-state analysis
3. Implications

Vowels, consonants and syllables

i		ɯi
e		o
	a	

Table 1: Monomoraic vowels and diphthongs

ii		ɯii
ee, ei		oo, oi
	aa, ai, ae	

Table 2: Bimoraic vowels and diphthongs

- Syllabic template: (C)V(V)(N)
- Syllabic prominence is determined by syllable weight and vowel height:
  - Syllable weight (Quantity & Coda): CVVN > CVV > CVN > CV
  - Vowel height (Quality): Low > Mid > High
- Onset of the syllable is irrelevant for the stress assignment.
- The only coda consonant in the syllable is nasals (e.g., [ŋ, n, m]).

Prominence scales

- Syllable weight and Vowel height interact differently between a foot-level and a word-level scale.

Ranking	IPA	Ranking	IPA
12	i	6	ii
11	e, o, ɯi	5	ee, ei, oo, oi, ɯii
10	a	4	aa, aV
9	iN	3	iiN
8	{e, o, ɯi}N	2	{ee, ei, oo, oi, ɯii}N
7	aN	1	{aa, aV}N

Table 3: A foot-level prominence scale: syllable weight is prioritized over vowel quality

Ranking	Foot-level Ranking	IPA	Ranking	Foot-level Ranking	IPA
12	12	i	6	6	ii
11	9	iN	5	3	iiN
10	11	e, o, ɯi	4	5	ee, ei, oo, oi, ɯii
9	8	{e, o, ɯi}N	3	2	{ee, ei, oo, oi, ɯii}N
8	10	a	2	4	aa, aV
7	7	aN	1	1	{aa, aV}N

Table 4: A word-level prominence scale: vowel quality is prioritized over syllable weight

Stress assignment

- Stress is assigned in a prosodic word '[$\cdots$]'.
- The stress placement is determined by the prominent syllable in each foot.

(1) a. [(à .tsi)(to. ká).kse].ro 'it crushed it'

b. [(à .wo)(te. hái).gʒi].ri 'we approached him/them'

- Stress is not assigned in the final syllable in a prosodic word.
- Possible foot types:

(2)

	prominence	pattern	type	pattern	type
a.	$\sigma_1 = \sigma_2$	$(\sigma_1\acute{\sigma}_2)$	`eqpromIamb`	$(\acute{\sigma}_1\sigma_2)$	`eqpromTrochee`
b.	$\sigma_1 < \sigma_2$	$(\sigma_1\acute{\sigma}_2)$	`goodIamb`		
c.	$\sigma_1 > \sigma_2$	$(\acute{\sigma}_1\sigma_2)$	`goodTrochee`		
d.	$\sigma_1 < \sigma_2$	$(\acute{\sigma}_1\sigma_2)$	`badTrochee`		

Rhythmic preference

- When the syllables are all in the **same prominence**, stress generally falls on the second syllable of each foot (`eqpromIamb`).

(3)

a.	o = o	[(o. gó).te].ro	'she will know it'
b.	i = i	[(i. rì)(pi. rí).ni.te]	'he will sit'

Stress clash avoidance

- Trochaic shift can occur in a foot with the same prominence as a stress clash-avoidance strategy if `eqpromTrochee` or `goodTrochee` follows it.

(4) *[(o.kò)(ri.kʃì)(tá.ka)] stress clash$_1$ due to stress avoidance in the ultima
*[(o.kò)(rì.kʃi)(tá.ka)] stress clash$_2$ due to clash avoidance
[(ò.ko)(rì.kʃi)(tá.ka)] trochaic stress due to clash avoidance
'she wore a nose-disc'

- However, trochaic shift can be **blocked** if the shift makes another stress clash context that is unresolved by trochaic shift (preceded by `goodIamb`).

(5) [(no.sà)(me.rè)(há.ka)] 'I got a blister'
*[(no.sà)(mè.re)(há.ka)]

Word-level stress

- Primary stress is on the **strongest** syllable.

(6)

a.	[(pi. ká)(bi.rì).ti]	'you get going'	a > i
b.	[(na. mám)(pi.jà).kse]	'I helped along'	$CV_iN > CV_i$
c.	[(nòo .ga).ka].ro	'I consumed it'	CVV > CV
d.	[(o.sà)(ráan .tai).ga.kse]	'they tore it...'	$CV_iVN > CV_iV$

- If two or more syllables tie for strongest in a prosodic word, the **rightmost** prominent syllable is stressed.

(7) [(i.rò)(gɯi. só).te] 'he will tie it' o = o

- Remaining prominent syllable in each foot becomes **secondary stress**.

Word types by footing patterns

- Stress placement depends on the number of syllables and footing pattern in a word, and each word type is **mutually exclusive**.

(8)

	footing pattern	word type
a.	$[(\sigma\sigma)^{+}(\sigma\sigma)\sigma]$	`Odd`
b.	$[(\sigma\sigma)^{+}(\sigma\sigma)]$	`EvenExhaustive`
c.	$[(\sigma\sigma)^{+}(\sigma\sigma)\sigma\sigma]$	`EvenNonExhaust`
d.	$[(\sigma\sigma)]$	`oneFoot`
e.	$[(\sigma\sigma)\sigma]$	`oneFootOneSyll`
f.	$[(\sigma\sigma)\sigma\sigma]$	`oneFootTwoSyll`

✍ '$[\cdots]$' marks the prosodic word edges and '$(\cdots)$' for foot boundaries.

✍ The superscript $^{+}$ indicates one or more repetitions.

Type 1: `Odd` $[(\sigma\sigma)^{+}(\sigma\sigma)\sigma]$

- The last syllable is unfooted in odd-number syllable words.
- Since the ultima is not stressed, **all foot types** except `badTrochee` can appear in the prosodic word.

(9)
a. [(o. kò)(wo. gó).te].ro 'she harvests it'
b. [(à .wo)(te. hái).gʒi].ri 'we approached him/them'
c. [(nò .ko)(gá .ko).ta].ro 'I asked about it'

Type 2: EvenExhaustive $[(\sigma\sigma)^{+}(\sigma\sigma)]$

- Exhaustive footing occurs if **the penult** is the **strongest** syllable in the prosodic word.

(10) a. $[(\text{i.kà.})(\text{ mán }\text{.ti})]$ ‘he told’

b. $[(\text{ò.gi.})(\text{ sá }\text{.kse})]$ ‘she lit a fire’

c. $[(\text{nò.be})(\text{ tá }\text{.kse})]\text{.ro}$ ‘I held onto it’

- The final foot must be either `eqpromTrochee` or `goodTrochee` to preserve the no stress in the ultima constraint.

Type 3: `EvenNonExhaust` $[(\sigma\sigma)(\sigma\sigma)^{+}\sigma\sigma]$

- The final two syllables are not footed, if the penult is **not strictly the strongest syllable** in the prosodic word.
- Regardless of the position of feet, **all foot types** except `badTrochee` can appear in the prosodic word.

(11)

a.	[(àn .ta)(mi. já).ta.kse]	'it (underbrush) grew up'
b.	[(kòn .te)(ja. tá).gze.ta]	'departed separately'
c.	[(i. rì .)(pi. rí).ni.te]	'he will sit'

Type 4: oneFoot [($\sigma\sigma$)]

- One-foot word must have **trochaic stress** (eqpromTrochee, goodTrochee, badTrochee), regardless of prominence.

(12) a. [(tá .ta.)] $\sigma_1 = \sigma_2$ 'what.INTERROG'

b. [(ʃí .gze.)] $\sigma_1 < \sigma_2$ 'run'

Type 5: `oneFootOneSyll` $[(\sigma\sigma)\sigma]$

- The last syllable is unfooted in odd-number syllable words.
- Since the ultima is not stressed, **any foot type** except `badTrochee` can appear in the prosodic word.
- The word with one foot followed by two unstressed syllable is independent from `Odd` $[(\sigma\sigma)(\sigma\sigma)^{+}\sigma]$, as the word-level prominence **overrides** the foot-level prominence.

(13)
a. [(o. gó).te].ro 'she will know it'
b. [(nóo .ga).ka].ro 'I consumed it'
c. [(im. pó).kse] 'he will come'

Type 6: `oneFootTwoSyll` $[(\sigma\sigma)\sigma\sigma]$

- The final two syllables are not footed, if the penult is **not strictly the strongest syllable** in the prosodic word.
- The word with one foot followed by two unstressed syllable is independent from `EvenNonExhaust` $[(\sigma\sigma)(\sigma\sigma)^{+}\sigma\sigma]$, as the word-level prominence **overrides** the foot-level prominence.

(14) a. [(ná .sin).tem.pa].ra FOOT: a < in, WORD: a > in

b. [(tsoŋ. ká).ta.ka] FOOT: oŋ > a, WORD: oŋ < a

Interim Summary

- The stress system of Nanti is 🔥**complicated**🔥.
- Six footing patterns are defined separately depending on:
  - the number of syllables
  - the number of feet in a word
  - penult prominence
- Based on the prominence relation in a foot, there are five foot types (e.g., `eqpromIamb`, `eqpromTrochee`...).
- In terms of word-level stress, primary stress can be assigned to:
  - the strongest syllable in a prosodic word OR
  - the rightmost prominent syllable if two or more syllables tie

Optimality Theory-based approach

- Crowhurst & Michael (2005): Optimality Theory (OT)-based account
  - stress placement as the result of ranked violable constraints
  - three interacting markedness scales: Quantity, Coda and Quality
- Lack of an exhaustive verification of the grammar
  - Crowhurst and Michael (2005) do not provide the full candidate space, so their chosen ranking does not verify the intended Nanti stress patterns across all possible inputs.

Computational approach

- Koser (2022): computational account for primary stress assignment
  - decomposition of patterns into two Strictly Piecewise functions: HEAVIEST and DESTRESS.

(15)

a.		b.		
o.ko.ri.kʃi.ta.ka		non.kan.tai.ga.kse		
↧		↧		HEAVIEST (L→R)
ó . kó .ri.kʃi. tá .ka		nón . kán . tái .ga.kse		
↧		↧		DESTRESS (L←R)
o.ko.rɪ.kʃɪ. tá .ka		non.kan. tái .ga.kse		

- The study only accounts for primary stress assignment.

Preliminaries

- The present study bridges the gap by offering a computational analysis that accounts for the metrical pattern in Nanti using the Language Toolkit (LTK; Lambert, 2024).
- It formalizes stress placement in Nanti as phonotactic patterns via a finite-state acceptor using Piecewise Local Expression Builder (Plebby) in LTK.

Preliminaries

- A 12-level prominence hierarchy: Syllable weight (4) $\times$ Vowel height (3)

IPA	Symbol	IPA	Symbol
i	A	ii	G
e, o, ɯi	B	ee, ei, oo, oi, ɯii	H
a	C	aa, aV	I
iN	D	iiN	J
{e, o, ɯi}N	E	{ee, ei, oo, oi, ɯii}N	K
aN	F	{aa, aV}N	L

Table 5: A foot-level prominence scale with syllabic representations.

IPA	Symbol	IPA	Symbol
i	A	ii	G
iN	D	iiN	J
e, o, ɯi	B	ee, ei, oo, oi, ɯii	H
{e, o, ɯi}N	E	{ee, ei, oo, oi, ɯii}N	K
a	C	aa, aV	I
aN	F	{aa, aV}N	L

Table 6: A word-level prominence scale with syllabic representations.

Preliminaries

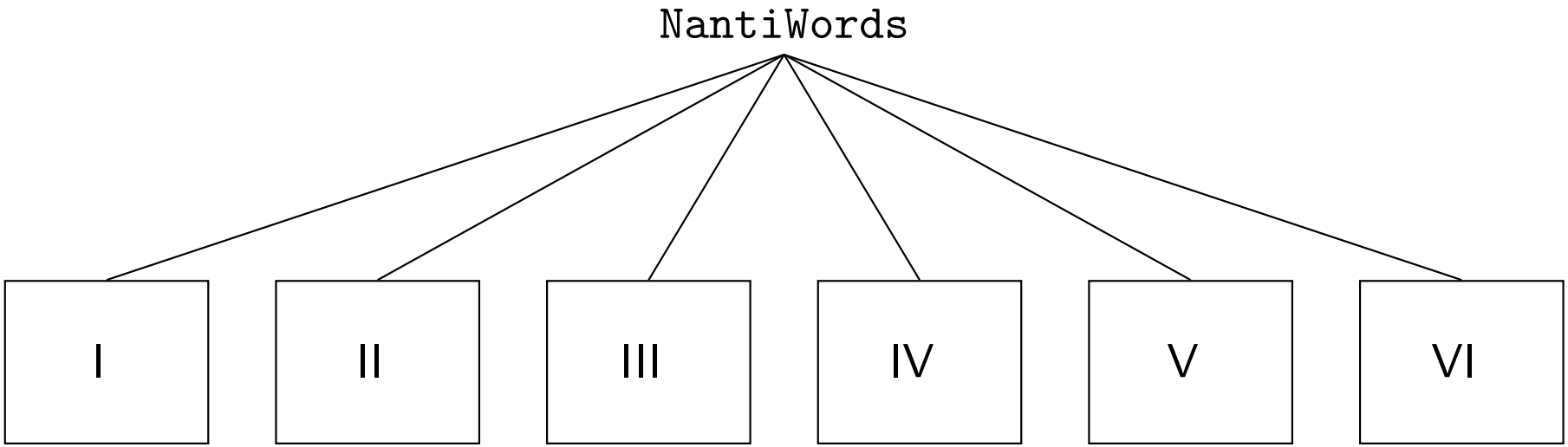
- ‘x’ (lower case) for unstressed; ‘X’ (upper case) for secondary stress; ‘Xs’ for primary stress
- ’P p’ for prosodic word edges

[(i.roo)(ga.ksem).pa].ro	P a K c Es a p b	‘he will have consumed it’
[(i.pì)(ri.ní).te]	P a A a As b p	‘he will sit’

Table 7: Examples of Nanti words with syllabic representations.

Analysis: Step 1

- The finite-state acceptor is a union of six disjoint word types.



- The different word types reflect the empirical generalizations: i) sensitive to syllable count, ii) exhaustive parsing, iii) penult sensitivity

Analysis: Step 1

- Each word shape is determined by the number of syllable and foot in a prosodic word, exhaustivitiy, and the prominence of the penultimate.
- Each type uniquely falls under one of six classes, mutually exclusive.

(16)

	footing pattern	word type
a.	$[(\sigma\sigma)^{+}(\sigma\sigma)\sigma]$	`Odd`
b.	$[(\sigma\sigma)^{+}(\sigma\sigma)]$	`EvenExhaustive`
c.	$[(\sigma\sigma)^{+}\sigma\sigma]$	`EvenNonExhaust`
d.	$[(\sigma\sigma)]$	`oneFoot`
e.	$[(\sigma\sigma)\ \sigma]$	`oneFootOneSyll`
f.	$[(\sigma\sigma)\sigma\sigma]$	`oneFootTwoSyll`

- Each word type is subject to various additional constraints (I-VI) to manage allowable footing types and their combinations.

Analysis: Step 1

- Each word type is defined as a combination of:
  - Word template
  - Primary stress constraints
  - Ending of the prosodic word (e.g., trochee, one or two unstressed syllables)
  - Constraints on equally prominent feet (`eqpromTrochee` & `eqpromIamb`)

Analysis: Step 2

- Based on these, `anyFeetSequence` are constructed as a word template with any foot combination.

(17)

	prominence	pattern	type	pattern	type
a.	$\sigma_1 = \sigma_2$	$(\sigma_1\acute{\sigma}_2)$	`eqpromIamb`	$(\acute{\sigma}_1\sigma_2)$	`eqpromTrochee`
b.	$\sigma_1 < \sigma_2$	$(\sigma_1\acute{\sigma}_2)$	`goodIamb`		
c.	$\sigma_1 > \sigma_2$	$(\acute{\sigma}_1\sigma_2)$	`goodTrochee`		
d.	$\sigma_1 < \sigma_2$	$(\acute{\sigma}_1\sigma_2)$	`badTrochee`		

(18) a. `FootTypes` = `goodIamb` | `eqpromIamb` | `goodTrochee` | `eqpromTrochee`

b. `anyFeetSequence` = $\texttt{FootTypes}^{+}$

(19) a. `trochee` = `goodTrochee` | `eqpromTrochee` | `badTrochee`

b. `iamb` = `goodIamb` | `eqpromIamb`

Distribution of foot types

- `goodIamb` and `goodTrochee` occur anywhere based on the prominence relation in each foot.
- `badTrochee` occurs only in `oneFoot`.
- `eqpromTrochee` and `eqpromIamb` appear in particular context to avoid stress clash.

The distribution of equally prominent feet

- These constraints are conjoined with `anyFeetSequence` to ban any illegal foot combinations.

(20) $\texttt{eqpromFootConstraint} = \neg\texttt{constraint1} \cap \neg\texttt{constraint2} \cdots \cap \neg\texttt{constraint6}$

constraints	configuration	example
`constraint1`	$\cdots$ `goodIamb` $\cdot$ `eqpromTrochee`$^{+}$ $\cdot$ `goodTrochee` $\cdots$	$\cdots$ (a B)(B b)(B a)$\cdots$
`constraint2`	`P` $\cdot$ `eqpromTrochee` $\cdot$ `goodIamb` $\cdots$	[(A a)(a B) $\cdots$
`constraint3`	`P` $\cdot$ `eqpromTrochee` $\cdot$ `eqpromIamb`$\cdots$	[(A a)(a A) $\cdots$
	`P` $\cdot$ `eqpromIamb` $\cdot$ `eqpromTrochee`$\cdots$	[(a A)(A a)$\cdots$
`constraint4`	`P` $\cdot$ `eqpromIamb` $\cdot$ `goodTrochee`$\cdots$	[(a A)(B a)$\cdots$
`constraint5`	`P` $\cdot$ `eqpromTrochee`$\cdots$	[(As a) $\cdots$
`constraint6`	$\cdots$ `goodIamb` $\cdot$ `eqpromTrochee` $\cdot$ `unstressed` $\cdot$ `p`	$\cdots$(a B)(A a) a]

Table 8: Constraints for legal foot combinations from `validwords`

Primary stress verification

- Stress is verified differently based on the number of syllables in a prosodic word.
- `culminativity`: exactly one primary stress in a single word

  (21) `*P a As b Bs a p`
- `hierarchy`: primary stress on the rightmost prominent syllable

  (22) `*P b Bs a A B b p`
- For odd-syllable words:
  - $\texttt{culminativity} \cap \texttt{hierarchy} = \texttt{wordStress_odd}$

Primary stress verification

- For even-syllable words:
  - `penult`: a boundary-anchored condition banning primary stress on a penultimate syllable, when a stressed syllable with the same prominence in the preceding foot

    (23) a. ⋯ `Cs a c c p`

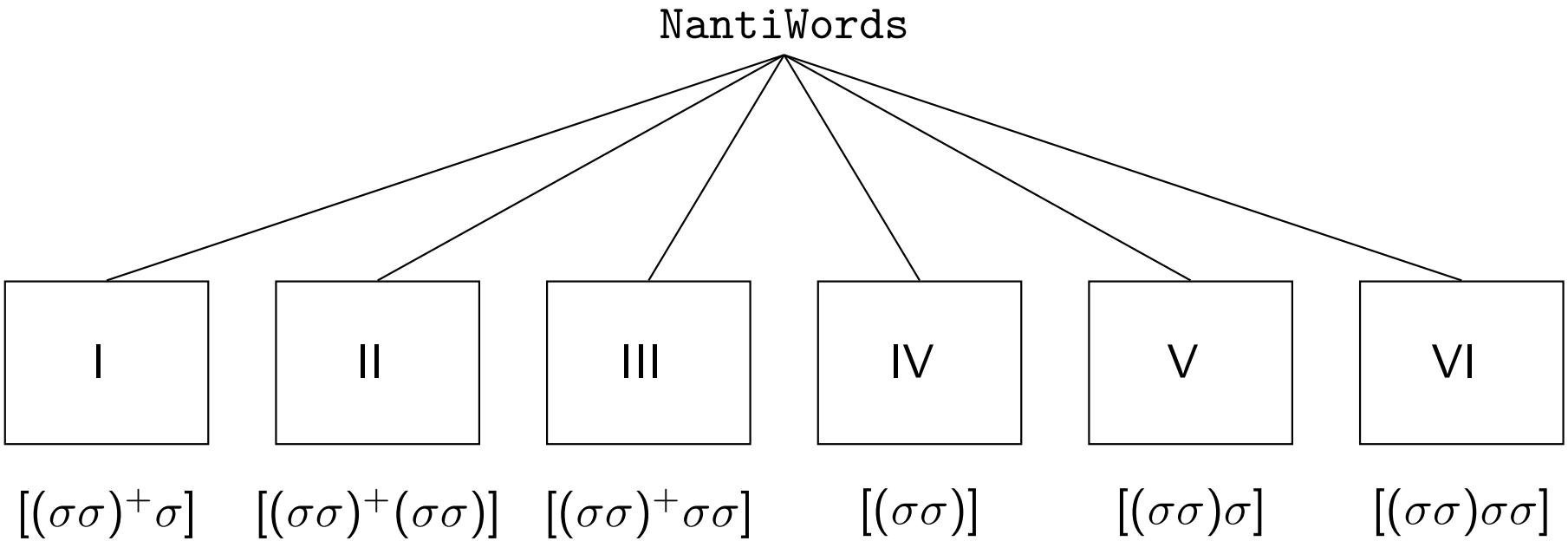
    b. ⋯ *`C a Cs c p`
  - `nopenultprom`: no unstressed syllable following the primary stress that outranks its prominence

    (24) a. ⋯ `A a Bs b p`

    b. ⋯ *`A a Bs c p`
  - `culminativity` ∩ `hierarchy` ∩ `nopenultprom` ∩ `penult`= `wordStress_even`

NantiWords

- All the "ingredients" are introduced to investigate how each word type is built.



(I) Odd $[(\sigma\sigma)^{+}(\sigma\sigma)\sigma]$

- validwords = anyFeetSequence ∩ ¬eqpromFootConstraint
- validwords_odd = validwords · unstressed
- Odd = validwords_odd ∩ wordStress_odd

<table>
<tr><td rowspan="3">word template
validwords_odd</td><td rowspan="3">validwords</td><td>any foot combination</td><td>anyFeetSequence</td></tr>
<tr><td>eqpromTrochee constraint</td><td rowspan="2">¬eqpromFootConstraint</td></tr>
<tr><td colspan="2">ending</td><td>one unstressed syllable</td></tr>
<tr><td colspan="3">Primary stress</td><td>wordStress_odd</td></tr>
</table>

(II) EvenExhaustive $[(\sigma\sigma)^{+}(\sigma\sigma)]$

- $\texttt{validwords_even} = \texttt{validwords} \cdot \texttt{trochee}$
- $\texttt{EvenExhaustive} = \texttt{validwords_even} \cap \texttt{wordStress_even}$

<table>
<tr><td rowspan="2">word
template
<code>validwords_even</code></td><td rowspan="2"><code>validwords</code></td><td>any foot combination
<code>eqpromTrochee</code>
constraint</td><td><code>anyFeetSequence</code></td></tr>
<tr><td><code>¬eqpromFootConstraint</code></td></tr>
<tr><td colspan="2">ending</td><td><code>trochee</code></td></tr>
<tr><td colspan="3">Primary stress</td><td><code>wordStress_even</code></td></tr>
</table>

(III) EvenNonExhaust $[(\sigma\sigma)^{+}(\sigma\sigma)\sigma\sigma]$

- validwords_even = validwords · unstressed · unstressed
- EvenNonExhaust = validwords_even ∩ wordStress_even

<table>
<tr><td rowspan="3">word
template
validwords_even</td><td rowspan="3">validwords</td><td>any foot combination</td><td>anyFeetSequence</td></tr>
<tr><td>eqpromTrochee
constraint</td><td rowspan="2">¬eqpromFootConstraint</td></tr>
<tr><td colspan="2">ending</td><td>two unstressed syllables</td></tr>
<tr><td colspan="3">Primary stress</td><td>wordStress_even</td></tr>
</table>

(IV) `onefoot`, (V) `oneFootOneSyll`, (VI) `oneFootTwoSyll`

IV	`oneFoot`	$[(\sigma\sigma)]$
V	`oneFootOneSyll`	$[(\sigma\sigma)\sigma]$
VI	`oneFootTwoSyll`	$[(\sigma\sigma)\sigma\sigma]$

- They use foot predicates independently built based on the word-level prominence scale.
- For IV , V and VI , the valid types are effectively listed since they are finitely many of these.

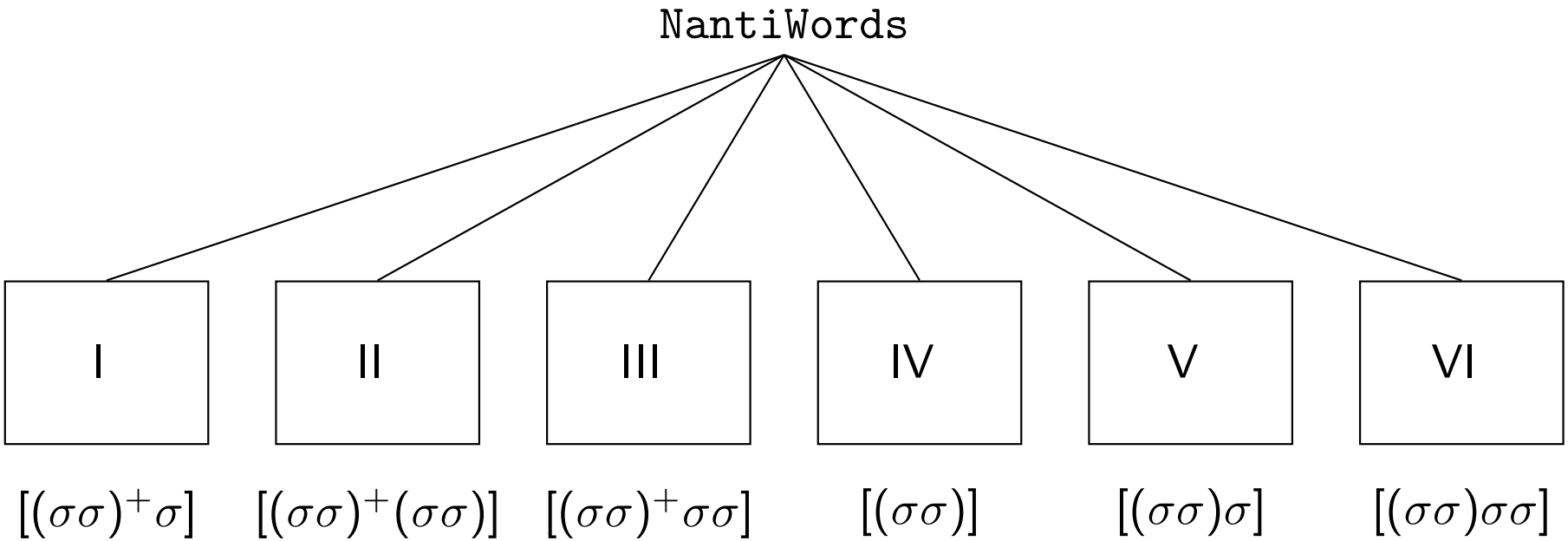
NantiWords

Branch	Template (@)	Foot level (∩)	Word level (∩)
Odd	$[(\sigma\sigma)^{+}(\sigma\sigma)\sigma]$	¬eqpromFootConstraint	wordStress_odd
EvenExhaustive	$[(\sigma\sigma)^{+}(\acute{\sigma}\sigma)]$	¬eqpromFootConstraint	wordStress_even
EvenNonExhaust	$[(\sigma\sigma)^{+}(\sigma\sigma)\sigma\sigma]$	¬eqpromFootConstraint	wordStress_even
oneFoot	$[(\acute{\sigma}\sigma)]$	n/a	trochee
oneFootOneSyll	$[(\sigma\sigma)\sigma]$	n/a	oddwords
oneFootTwoSyll	$[(\sigma\sigma)\sigma\sigma]$	n/a	evenwords

Table 9: Composition of the acceptor NantiWords.

NantiWords

- Each word type is unioned as NantiWords.



Results

- For practicality, the study examined 3-level (`A-C`) and 5-level (`A-E`) acceptors.
  - 3-level acceptor: for computing which subregular class it belongs to
  - 5-level acceptor: for verifying the words from Crowhurst and Michael (2005)
- It was necessary to look at 5-level machine due to the two different prominence scales for foot and word.

levels	3	4	5	6	7	8	$\cdots$	12
# of states	215	450	632	821	1,036	1,276	$\cdots$	$\approx$ 2,420

Table 10: The attested number of states of the acceptor in each level.

Results

- All words with five levels extracted from Crowhurst & Michael (2005) (49 words out of 113 words) were tested (`truewords`):

```
:subset truewords NantiWords
> true
```

- These words were also manipulated to put stress in the wrong position (`falsewords`):

```
:subset falsewords ¬NantiWords
> true
```

Results

- The program *Classify* in LTK identified the language's class membership as **First Order definable with precedence ($<$)**.

Successor	Precedence
Monadic Second Order	
	First Order ($<$) ✓
First Order ($\lhd$) ✗	First Order 2 ($<$) ✗
Locally Testable ✗	Piecewise Testable ✗
Strictly Local ✗	Strictly Piecewise ✗

12-level abstraction and its computational complexity

- The complexity exceeds that of other stress patterns in StressTyp2 (Goedemans et al., 2015), which Lambert (2026) found to be propositional with Tier-based Locally Testable and Piecewise Testable factors.
  - the most complicated system in StressTyp2 has five syllable types and at most 40 states
- Adding a single level produces a bigger growth in the acceptor.
- As Koser (2022) notes, the prominence richness does not increase computational complexity.
- The number of states increases because more states are needed to keep track of the distinctions between the prominence levels, but the logic of the machine is the same.

Learnability consequences

	CNL		*Propositional*		*First-Order*	
	SL	SP	LT	PT	FO($\triangleleft$)	FO($<$)
`validwords`	`false`	`false`	`false`	`false`	?	`true`
`penult`	`false`	`false`	`false`	`false`	`false`	`true`
`nopenultprom`	`false`	`true`	`false`	`true`	`false`	`true`
`wordLevelStress`	`false`	`true`	`false`	`true`	`false`	`true`
`culminativity`	`false`	`true`	`false`	`true`	`true`	`true`
`hierarchy`	`false`	`true`	`false`	`true`	`false`	`true`

Table 11: Subregular results of class membership of each grammar in `NantiWords`.

✍ The result for FO($\triangleleft$) in `validwords` failed to return the result; presumably, it could be considered as `false`.

Learnability consequences

- Most of the predicates in the grammar remain within the propositional region.
- However, `penult` and `validwords` fall outside this region, placing it outside more easily learnable subregular regions.
- `penult` is computed based on the right edge of the prosodic word, and SP or PT cannot calculate the penult position.
- Since `validwords` looks at the grammatical footing combinations in a word, it needs to be computed non-locally.

Conclusion

- The study provided a finite-state analysis of Nanti's stress system.
- The grammar decomposed the patterns into formal predicates over symbolic representations.
- The finite state acceptor caputured the attested words and classified them as an FO-definable with precedence, but nothing weaker.

Conclusion

- Future directions
  - Further verification of other Nanti words beyond A-E by reducing levels
  - A more complete logical analysis using the program *Factorize* in LTK
  - Test other languages with complicated stress patterns
  - Investigating the learnability of Nanti